Western Blotting Protocol

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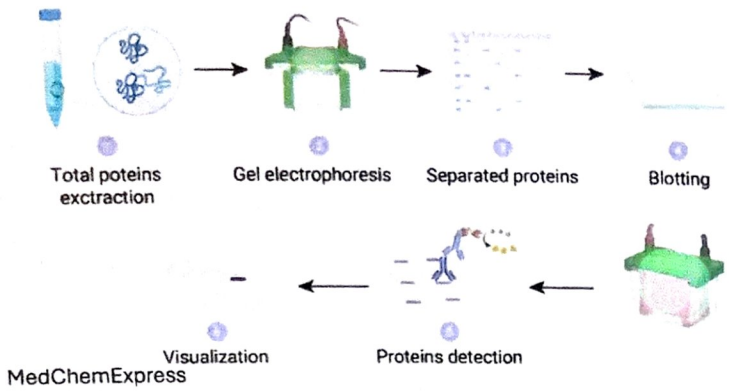
Safety statement: Prior to beginning an experiment, please read the Standard Operating Procedure (SOP) for this protocol as well as the Safety Data Sheets (SDS) for all chemical reagents listed. The SOP will contain a list of all chemicals for this protocol as well as directions for handling and disposal of hazardous wastes. It will also contain proper PPE guidelines and emergency protocols for accidents involving hazardous chemicals. Always be aware of your surroundings in the lab and be sure to communicate any hazards to those around you when performing an experiment.

Hazard statement: Any hazardous chemicals handled within this protocol will be marked with the following notation: (H). Please refer to the associated SOP to better understand the hazards and how to work with and dispose of these chemicals safely.

Background:

"A western blot experiment, or western blotting (also called immunoblotting), was introduced by Towbin, et al. in 1979 and is now a routine technique for protein analysis. The specificity of the antibody-antigen interaction enables a target protein to be identified in the midst of a complex protein mixture. Western blotting can produce qualitative and semi-quantitative data about the protein of interest. The first step in a western blotting procedure is to separate the macromolecules in a sample using gel electrophoresis. Subsequently, the separated molecules are transferred or blotted onto a second matrix, generally a nitrocellulose or polyvinylidene difluoride (PVDF) membrane. Next, the membrane is blocked to prevent any nonspecific binding of antibodies to the surface of the membrane. Most commonly, the transferred protein is then probed with a combination of antibodies: one antibody specific to the protein of interest (primary antibody) and another antibody specific to the host species of the primary antibody (secondary antibody). Often the secondary antibody is complexed with an enzyme, which when combined with an appropriate substrate, will produce a detectable signal" (ThermoFisher).

Western Blotting Workflow



Equipment:

1. Centrifuge
2. Plastic gel support
3. Rubber stoppers
4. Glass gel casts
5. Cast clips
6. Gel combs
7. Gel box
8. Electrodes
9. Plastic trays
10. Transfer sandwich cases
11. Sponges
12. Blotting paper
13. Nitrocellulose membranes
14. Magnetic plate + stir bar
15. Heating block for Eppendorf tubes
16. Sonicator
17. Blot imager
18. Pipettor + gel-loading pipette tips

Reagents:

1. Trypsin-EDTA (0.05%)
2. 2X Laemmli buffer
3. 1M dithiothreitol (DTT)
4. 10X Tris-Glycine-SDS (TGS)
5. 10X Tris-Glycine (TG)
6. Stock gel components:
    - Deionized water (diH_2O)
    - 30% Acrylamide (H)

- Tris-HCl (0.5M **and** 1.5M)
- 10% sodium dodecyl sulfate (SDS) (H)
- Ammonium persulfate (APS) (H)
- Tetramethylethylenediamine (TEMED) (H)

7. Pre-stained molecular weight standards (i.e. ladder)
8. PonceauS stain
9. Coomassie brilliant blue dye
10. 1X PBS + 0.1% Tween-20 (PBST)
11. Dried, nonfat milk
12. ECL (**e**nhanced **c**hemi**l**uminescence) reagents: stable peroxide and luminol
13. 100% methanol (H)
14. Primary detection antibody
15. Secondary antibody cross-adsorbed to horseradish peroxidase (HRP)

Procedures:

Preparing samples:

1. To prepare sample buffer: dilute DTT to a final concentration of 100mM in pre-formulated 2X Laemmli buffer.
    a. Example: For 1mL total volume, mix 100µL of 1M DTT with 900µL of 2X Laemmli buffer.
2. Harvest cells by trypsinization and centrifuge to pellet.
3. Resuspend pelleted cells in appropriate volume of 1X PBS. Remove small aliquot for counting.
4. Count cells *twice* on the Countess automated cell counter and average the results.
5. Centrifuge cells again to pellet.
6. Based on the cell counts, add an appropriate volume of PBS to pelleted cells and resuspend.
7. Add an equivalent volume of 2X Laemmli sample buffer and triturate the sample until the sample buffer has lysed the cells.
    a. For most proteins, lysates should be in a range between 50,000-100,000 cells per 10µL of lysate.
    b. Freeze samples at -20°C if not using immediately.
8. Sonicate samples using a probe sonicator at 30% amplitude for 3-5 seconds. Make sure the sample froths to ensure complete shearing of genomic DNA.
9. Snap-spin tubes to bring liquid to the bottoms of the tubes.
10. Heat samples for 10 minutes at 75°C in a heating block. Cool for 1 minute before loading into the wells of a polyacrylamide gel.

Casting polyacrylamide gels:

1. Assemble the gel cast as follows:
    a. Choose either 0.75mm, 1.0mm or 1.5mm glass casts. The wider the gap, the more gel solution is needed to fill the cast to the appropriate level.
    b. Secure the glasses into the cast holder.
    c. Place rubber stoppers in the appropriate places of the plastic support.
    d. Put the assembled cast onto the rubber stoppers and clip the cast into place.

e. To test whether a seal has been formed between the bottom of the cast and the rubber, use diH_2O. If the level has moved down after a short time, the seal is broken and will have to be adjusted. ALWAYS CHECK SEAL BEFORE PUTTING GEL SOLUTION INTO THE CAST.

2. Once the seal is intact, make the solution for the *SEPARATING* gel. Use the following table:

For 2 gels, 0.75mm

	Stacking (4%)	5%	7%	10%	15%
Water	3 mL	5.67 mL	5 mL	4 mL	2.34 mL
30% Acrylamide	667 µL	1.67 mL	2.33 mL	3.33 mL	5 mL
1.5 M Tris-HCl, pH 8.8	0	2.5 mL	2.5 mL	2.5 mL	2.5 mL
0.5 M Tris-HCl, pH 6.8	1.25 mL	0	0	0	0
10% SDS	50 µL	100 µL	100 µL	100 µL	100 µL
10% APS	30 µL	60 µL	60 µL	60 µL	60 µL
TEMED	5 µL	5 µL	5 µL	5 µL	5 µL

For 2 gels, 1.5mm

	Stacking (4%)	5%	7%	10%	15%
Water	6 mL		10 mL	8 mL	
30% Acrylamide	1.334 mL		4.66 mL	6.66 mL	
1.5 M Tris-HCl, pH 8.8	0		5 mL	5 mL	
0.5 M Tris-HCl, pH 6.8	2.5 mL		0	0	
10% SDS	100 µL		200 µL	200 µL	
10% APS	60 µL		120 µL	120 µL	
TEMED	10 µL		10 µL	10 µL	

3. Fill the cast with the separating gel solution using a pipette. Layer 100% isopropanol over the gel solution in the cast to prevent evaporation. Let polymerize for ~40 minutes.
   a. Once a clear line forms between the gel solution and the alcohol, the gel has solidified and the alcohol can be collected in a plastic container.
   b. Rinse the top of the gel with diH_2O and use a KimWipe to dry the space prior to filling the stacking gel.
4. Now make the *STACKING* gel (always 4% polyacrylamide; refer to table above)

... the cast the rest of the way with stacking gel and immediately place well combs into the stacking gel.

6. After ~30 minutes, the gels are ready for use in electrophoresis.
    a. If not using immediately, wrap gels with the combs still inserted in paper towels soaked in diH_2O. Wrap tightly in plastic wrap and lay flat at 4°C.

Running the gel:

1. Prepare the Running buffer in a graduated cylinder:
    a. 100mL 10X TGS
    b. 900mL diH_2O
2. Assemble the gel box:
    a. Secure the gels into the holders and test them. They should not be loose at all.
    b. Place this apparatus into the gel box and fill the space between the gels with Running buffer. Make sure the buffer does not leak from between the gels.
    c. Fill the rest of the box with Running buffer so that it is level with the buffer in between the gels.
    d. Pull the combs straight up and out without disturbing the wells. Orient your eyes to see the wells before loading the samples.
3. Load samples into wells; first lane is always a molecular weight standard.
    a. 10-30µL for 10-well gel.
    b. 5-15µL for 15-well gel.
4. Cover gel box with lid, making sure to line up the electrodes correctly (red-red; black-black)
5. Plug lid into power source and run gel at 100V constant voltage for 10-20 minutes to move samples through the stacking gel.
6. Increase voltage to 150V and run for approximately 45 min – 1 hour.
7. Once the dye front has run completely off, turn off power source, disconnect the wires, and open carefully.
8. Remove the gel and disassemble:
    a. Crack open the cast using a metal spatula.
    b. Cut off stacking gel and wells with a gel knife.
9. The gel is now ready to be transferred to a membrane.

Transferring proteins to nitrocellulose membrane:

1. Prepare Transfer buffer ***beforehand***:
    a. 700mL diH_2O
    b. 100mL 10X Tris-glycine
    c. 200mL 100% methanol
    d. Mix and refrigerate to chill (or place in ice bath to chill quickly)
2. Assemble the transfer apparatus (box with electrodes oriented in the midline and an icepack), and place in -20°C freezer **at least 90 minutes** prior to transfer.
3. Set up your transfer "sandwich" in a plastic tray submerged in Transfer buffer. From bottom to top:
    a. Black side of sandwich case
    b. Sponge

*Use a roller to gently push air bubbles out after placing these components. Start the roller from the center and roll slowly outward in both directions.

c. 2 pieces of blotting paper
d. Gel
e. Nitrocellulose membrane*
f. 2 pieces of blotting paper*
g. Sponge
h. Transparent side of sandwich case

4. Close the "sandwich" and squeeze tight. Lock it closed with the sandwich case attachment.
5. Retrieve the transfer apparatus from the freezer and place within a plastic tray on a magnetic plate.
6. Slide your "sandwich" into the transfer apparatus (oriented with the BLACK side of the SANDWICH CASE facing the BLACK wall of the ELECTRODE SUPPORT).
7. Fill the apparatus with ice cold Transfer buffer to the top and place a magnetic stir bar and set to ~250-300rpm.
8. Place the lid onto the apparatus, making sure to match electrodes.
9. Turn on the power and switch to "Constant Current." Set to 250mAmp and run the transfer for 1 hour – 1 h 20m.

Analyzing transfer efficiency:

1. Following transfer, disassemble the apparatus, open your "sandwich," and wash the membrane with PBST to remove residual methanol.
2. Wash gel in diH_2O to remove residual methanol.
3. To analyze transfer efficiency, two different stains are utilized:
    a. Coomassie brilliant blue
        i. Stain **GEL** for 10 minutes with gentle rocking.
        ii. Replace liquid in bottle, then wash excess Coomassie out of gel with diH_2O for 10 minutes/wash until bands are clearly visible (can leave overnight if needed).
        iii. Image as needed.
    b. PonceauS (0.5g PonceauS dissolved in 5% acetic acid in diH_2O)
        i. Stain **MEMBRANE** for 5 minutes with gentle rocking.
        ii. Replace PonceauS in bottle, then wash excess away with diH_2O until bands are clearly visible.
        iii. Image as needed.

Detecting protein(s) of interest:

1. Following PonceauS staining, wash away residual stain with PBST.
2. Make block buffer as follows:
    a. Weigh out 2.5g dried, nonfat milk and place in a 50mL centrifuge tube.
    b. Add 25mL PBST to the tube and vortex until milk dissolves.
    c. Add another 25mL PBST to make 50mL of block buffer (i.e. 5% w/v nonfat milk).
3. Pipette 15mL block buffer onto membrane and rock gently for 1 hour at room temperature OR overnight at 4°C.

STOP

4. Remove block buffer and wash membrane 3X with PBST (3 min/wash).

5. To make *primary* antibody solution, dilute milk to 0.5% in PBST and then add antibody at the correct dilution.
    a. If using a new antibody, make sure to test serial dilutions prior to your experiment to ensure the use of an optimal dilution for sample detection.
6. Rock gently for 1 hour at room temperature OR overnight at 4°C.
7. Replace primary antibody solution in tube and store. **This is good for about 1 week at 4°C**. Make sure to whiff the solution if you suspect the milk has spoiled.
8. Wash the membrane 3X with PBST (3 min/wash).
9. To make *secondary* antibody solution, dilute milk to 0.5% in PBST and then add antibody at the correct dilution.
    a. It is important that the species in which the secondary antibody was raised against **matches** the species in which the primary antibody was raised (e.g. your primary antibody is raised in mouse, so therefore your secondary antibody must be raised *against* mouse in order for you to detect your sample)
10. Rock gently for 1 hour at room temperature.
    a. You CANNOT incubate overnight. This will lead to oversaturation of secondary antibody on the blot and consequently an increase in background when imaging.
11. Remove secondary antibody solution and wash 3X with PBST (3 min/wash).
12. Wash membrane once for 1 minute with diH_2O.
13. Using tweezers, pick up the blot, wick off excess liquid and carefully lay it down on an imaging tray. Make sure the blot is completely flat on the tray and does not have any air bubbles underneath as this can lead to background when imaging.
14. Mix stable peroxide and luminol in equal volumes to create your ECL detection solution.
15. Pipette ECL solution on top of blot and spread solution over entire blot.
16. Place tray immediately in LiCor imaging station receptacle and close the machine.
17. Open the ImageStudio software and set the exposure times for your chemiluminescence channel as well as your 700nm channel (used to visualize the molecular weight standards).
18. Once images are done developing, save them as TIFF files with the highest resolution available.

Quantification of bands:

1. ImageJ
    a. Open TIFF file within the ImageJ software.
    b. Crop your image as needed and save as a separate file.
    c. Create your region of interest (ROI) around the largest band of your blot.
        i. You can save the dimensions of your ROI in the case that you accidentally lose the dimensions.
    d. Press 'Ctrl+M' to take a measurement of the band intensity. A new window should open within ImageJ that records the measurements.
    e. Keeping dimensions of your ROI consistent, take measurements of all bands on the blot as well as a background measurement.

f. Copy measurements to an Excel worksheet where statistical analysis can be performed.

ImageStudio

a. Within the same software used to image, go to the 'Analysis' tab.
b. Press the 'Add rectangle' button and select your largest band. This creates a custom-fitted box around the band, no manual sizing needed.
c. Copy and paste that box around all of the bands on your blot. There is no need to take a background measurement on this software since it is programmed to automatically remove background.
d. Values appear either under or above each box. These are the **background-corrected** intensity values.
e. Copy these values to an Excel worksheet where statistical analysis can be performed.